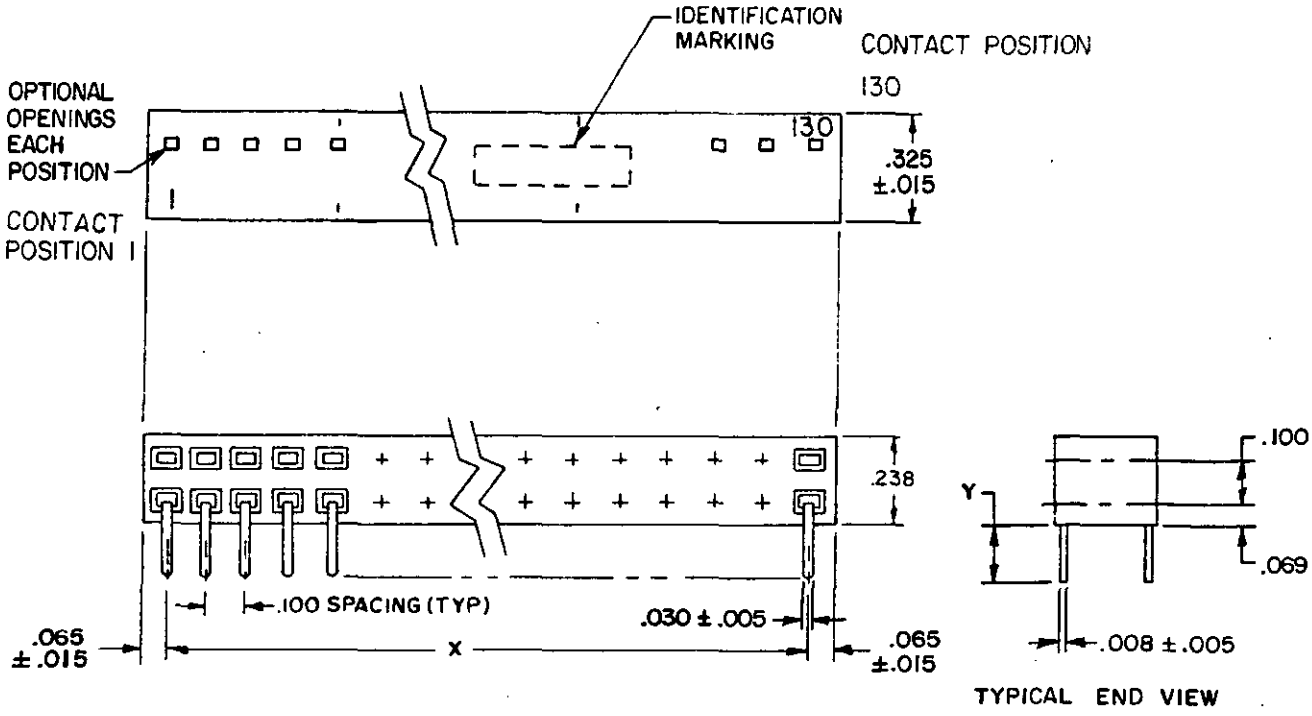
MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: RECEPTACLE, DOUBLE ROW, 4 THRU 130 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (1/16", 3/32", AND 1/8") (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.

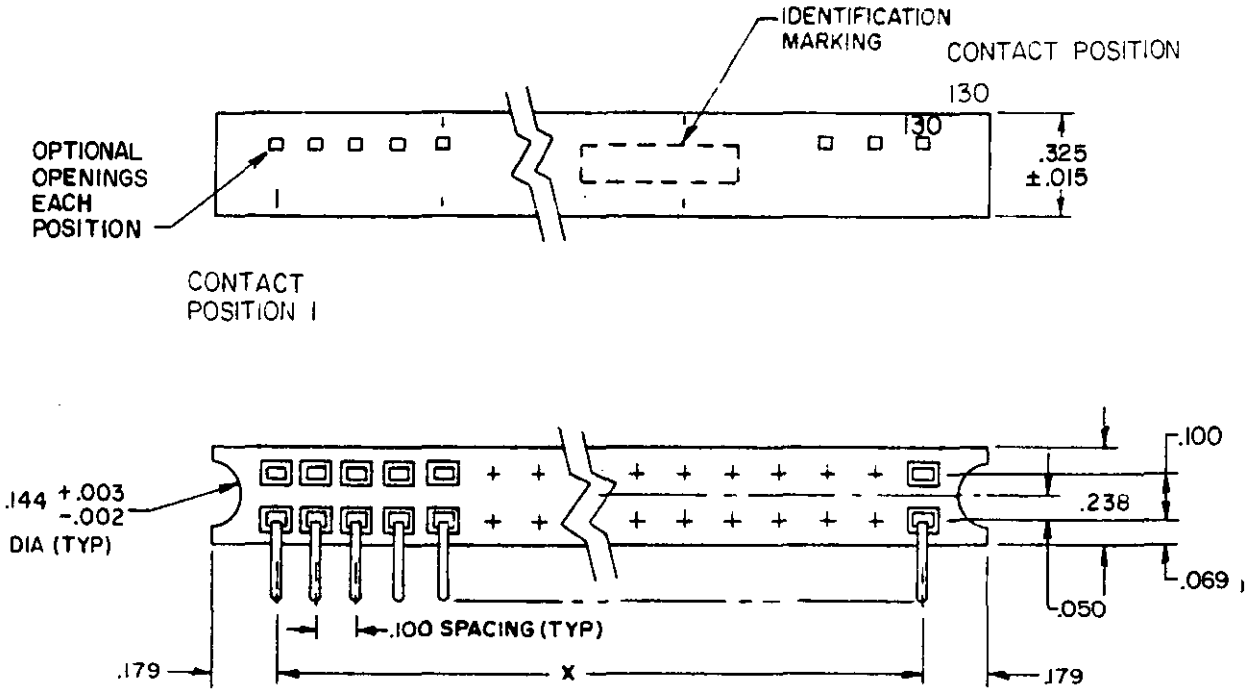


HOUSING CONFIGURATION 1

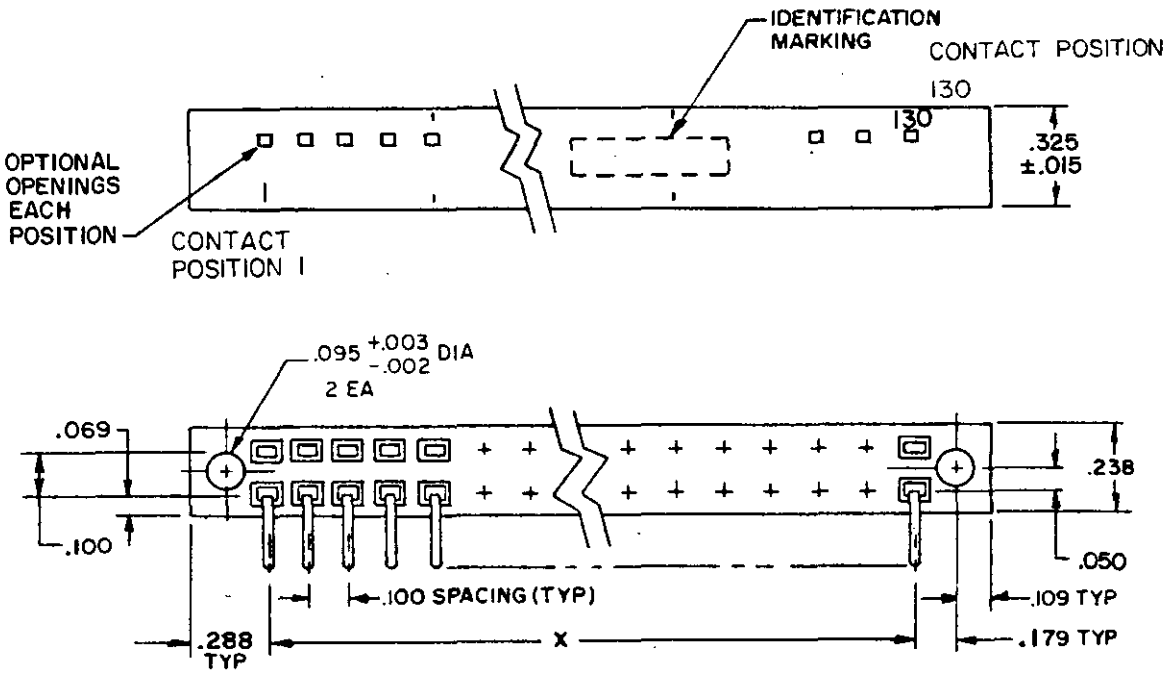
FIGURE 1. Connectors, receptacle, .100 spacing.

Ⓐ denotes changes.

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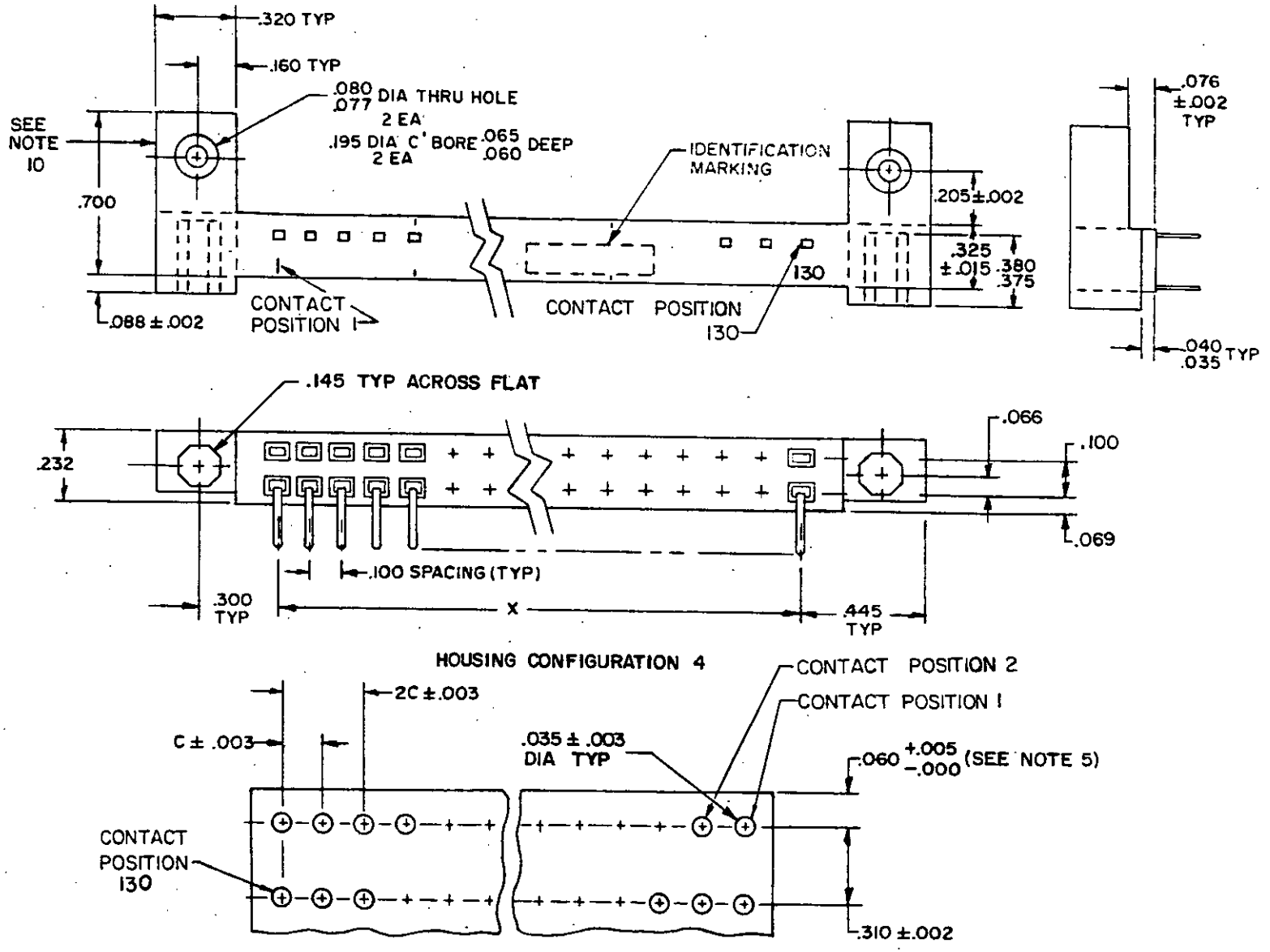
HOUSING CONFIGURATION 2



HOUSING CONFIGURATION 3

FIGURE 1. Connectors, receptacle, .100 spacing - Continued.

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RECOMMENDED PRINTED CIRCUIT BOARD HOLE LAYOUT

C - Contact centerline to be .100 (2.54 mm); ±.003 (.08 mm) tolerances not to accumulate within one connector pattern.

FIGURE 1. Connectors, receptacle, .100 spacing - Continued.

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INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.076	1.93	.144	3.66	.205	5.21
.003	.08	.077	1.96	.145	3.68	.232	5.89
.005	.13	.080	2.03	.160	4.06	.238	6.05
.015	.38	.088	2.24	.179	4.55	.288	7.32
.035	.89	.095	2.41	.195	4.95	.300	7.62
.040	1.02	.100	2.54			.310	7.87
.050	1.27					.320	8.13
.060	1.52					.325	8.26
.065	1.65					.375	9.52
.066	1.68					.380	9.65
.069	1.75					.445	11.30
						.700	17.78

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is ±.010 (.25 mm).
4. The mating post array must be maintained on .100 centerline grid spacing.
5. Post engagement length is .323 maximum, .308 minimum, as measured from reference datum, when connector is mounted per figure 1. Total mating positions to be determined by connector size per table I.
6. Post location tolerance not to accumulate.
7. Mating post tips must be maintained within .015 diameter true position.
8. Mating posts may consist of posts staked directly into printed circuit boards, or mounted in insulator bodies.
9. The first and last circuit shall be numbered and every fifth circuit shall have an identifiable mark on the top surface of the housing beginning with the first circuit on the left (#1) locking at the mating face of the connector.
10. Mounting ears shall be molded or bonded to the insulator.
11. Housing configurations 1, 3, and 4 terminate directly to the board and therefore do not meet the requirements of MIL- STD-275.
(A) 12. Housing configuration 2 mates with M55302/165 and does meet the requirements of MIL-STD-275 for two piece connectors.

FIGURE 1. Connectors, receptacle, .100 spacing - Continued.

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TABLE I. Dimensions. 1/

Code	Number of contacts	Dimension X	Code	Number of contacts	Dimension X	Code	Number of contacts	Dimension X	Code	Number of contacts	Dimension X
AA	4	.100 (2.54)	BA	36	1.700 (43.18)	CA	68	3.300 (83.82)	DA	100	4.900 (124.46)
AB	6	.200 (5.08)	BB	38	1.800 (45.72)	CB	70	3.400 (86.36)	DB	102	5.000 (127.00)
AC	8	.300 (7.62)	BC	40	1.900 (48.26)	CC	72	3.500 (88.90)	DC	104	5.100 (129.54)
AD	10	.400 (10.16)	BD	42	2.000 (50.80)	CD	74	3.600 (91.44)	DD	106	5.200 (132.08)
AE	12	.500 (12.70)	BE	44	2.100 (53.34)	CE	76	3.700 (93.98)	DE	108	5.300 (134.62)
AF	14	.600 (15.24)	BF	46	2.200 (55.88)	CF	78	3.800 (96.52)	DF	110	5.400 (137.16)
AG	16	.700 (17.78)	BG	48	2.300 (58.42)	CG	80	3.900 (99.06)	DG	112	5.500 (139.70)
AH	18	.800 (20.32)	BH	50	2.400 (60.96)	CH	82	4.000 (101.60)	DH	114	5.600 (142.24)
AJ	20	.900 (22.86)	BJ	52	2.500 (63.50)	CJ	84	4.100 (104.14)	DJ	116	5.700 (144.78)
AK	22	1.000 (25.40)	BK	54	2.600 (66.04)	CK	86	4.200 (106.68)	DK	118	5.800 (147.32)
AL	24	1.100 (27.94)	BL	56	2.700 (68.58)	CL	88	4.300 (109.22)	DL	120	5.900 (149.86)
AM	26	1.200 (30.48)	BM	58	2.800 (71.12)	CM	90	4.400 (111.76)	DM	122	6.000 (152.40)
AN	28	1.300 (33.02)	BN	60	2.900 (73.66)	CN	92	4.500 (114.30)	DN	124	6.100 (154.94)
AP	30	1.400 (35.56)	BP	62	3.000 (76.20)	CP	94	4.600 (116.84)	DP	126	6.200 (157.48)
AQ	32	1.500 (38.10)	BQ	64	3.100 (78.74)	CQ	96	4.700 (119.38)	DQ	128	6.300 (160.02)
AR	34	1.600 (40.64)	BR	66	3.200 (81.28)	CR	98	4.800 (121.92)	DR	130	6.400 (162.56)

1/ Metric equivalents are in parentheses.

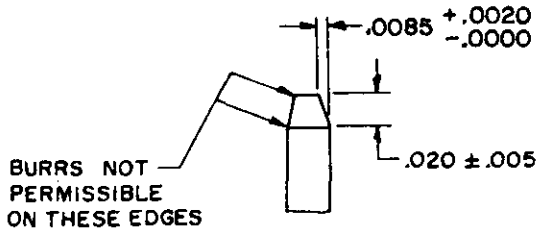
FIGURE 1. Connectors, receptacle, .100 spacing - Continued.

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Table II. Printed circuit board and plating thickness

Code	Board thickness	Y Dimension	Contact area plating thickness	Remainder plating thickness
A	.062	.115	Class 1	Flash
B	.062	.115	Class 1	Class 1
C	.093	.145	Class 1	Flash
D	.093	.145	Class 1	Class 1
E	.125	.175	Class 1	Flash
F	.125	.175	Class 1	Class 1

FIGURE 1. Connectors, receptacle, .100 spacing - Continued.



INCHES	MM
.0020	.051
.005	.13
.0085	.216
.020	.51

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 2. Mating post tip configuration.

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REQUIREMENTS:

Dimensions and configurations: See figures 1 and 2 and tables I and II.

Material: In accordance with MIL-C-55302.

Plating: The contact plating shall be gold per MIL-G-45204, type II, grade C, class 1 as specified in table II, over nickel plating per QQ-N-290, class 2, 30 to 150 microinches.

Contact cavities: See figure 1.

Printed wiring terminations: See figure 1.

Contact identification: See figure 1.

Polarization: Connector polarization may be accomplished by use of a keying plug (P/N M55302/127-00KY) which is inserted into the particular contact circuit to be keyed. Keying plug is nylon per MIL-M-20693.

Color: Natural.

Alinement: To be accomplished by user.

Method of mounting: Not applicable.

Oversized pin exclusion (sockets only): Shall not accept a .050 inch diameter pin.

Contact engagement and separation forces:

Engagement force: Shall be 6 ounces maximum after three insertions of a .0260 +.0000, -.0001 inch square pin.

Separation force: Shall be .75 ounce minimum.

Ⓐ Mating and unmating: The maximum connector mating force shall be 7 ounces times the number of contact positions, and the minimum connector unmating force shall be .75 ounce times the number of contact positions. For purposes of testing, the mating plug connector shall be considered to be the correct number of .025 wrappost in accordance with MIL-STD-1130, gold-plated. Maximum post engagement length is .323 inch and minimum post engagement length is .308 inch, when connector is mounted as shown on figure 1.

Mating post: See figure 2. For total mating positions, see table I.

Contact rating: 3.0 amperes maximum.

Contact resistance: No individual contact pair shall have a resistance exceeding 12 milliohms when mated to a gold-plated .025 wrappost in accordance with MIL-STD-1130.

Contact retention: Not applicable.

Dielectric withstanding voltage:

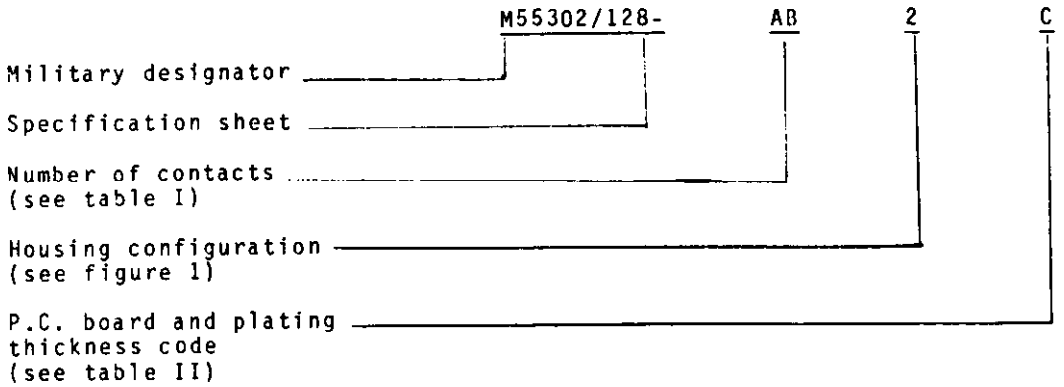
Sea level: 750 V rms.

High altitude: 275 V rms.

Contact bend test: The body of the contact shall be slowly bent through 90° and then returned to the original position. A bend through 90° and return to original position shall be defined as one bend. The test shall include three bending cycles.

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Part number: Shall be as shown in the following example:



Custodian:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - AS, OS, SH
Air Force - 11, 85
DLA - ES

User activities:
Army - AT, AV, ME
Navy - MC
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Air Force - 17

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